

# MANISH MAN SHRESTHA

Firmware & Embedded Systems Engineer | IoT Developer

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## Hiring Team

Motion Recruitment Partners  
Lake Forest, CA

## Re: Sr. Embedded Systems Engineer (Lake Forest, CA)

Dear Hiring Manager,

I am excited to apply for the Sr. Embedded Systems Engineer position. With 15 years building real-time embedded firmware - and a focus on low-power, wireless IoT systems carried from concept through production - this role maps closely to what I do best.

Here is how my experience lines up with what the role calls for:

- **End-to-end embedded development (conception to production):** As a firmware engineer I delivered firmware for four distinct commercial products, and as a CTO I owned embedded systems from requirements and architecture through coding, test, debug, and production release.
- **C/C++/Python and RTOS:** I develop real-time firmware in Embedded C/C++ - with Python for tooling and analysis - on ARM Cortex-M and ESP32 using FreeRTOS, including device drivers, ISR design, and task scheduling under hard timing constraints.
- **Low-power wireless and IoT:** I design distributed ESP32-S3 IoT nodes with Wi-Fi/BLE and secure MQTT/TLS telemetry - including a 120-day field deployment that ran with zero failures - and routinely measure and optimize power consumption and firmware footprint on resource-constrained MCUs.
- **Signal processing (RF-adjacent):** I have implemented FFT, IIR, and FIR algorithms directly on ARM Cortex-M4 for real-time, on-device processing - a solid foundation for communications and RF-oriented work.
- **CI/CD and collaboration:** I established Git-based version control and YAML/Docker CI/CD pipelines for repeatable firmware builds and release management, and I work effectively across QA, hardware, and design teams.

I am based in Brookings, SD and am glad to relocate to Lake Forest for an on-site role. I would welcome the chance to discuss how my low-power wireless firmware experience can contribute to your team. Thank you for your consideration.

Sincerely,

**Manish Man Shrestha**